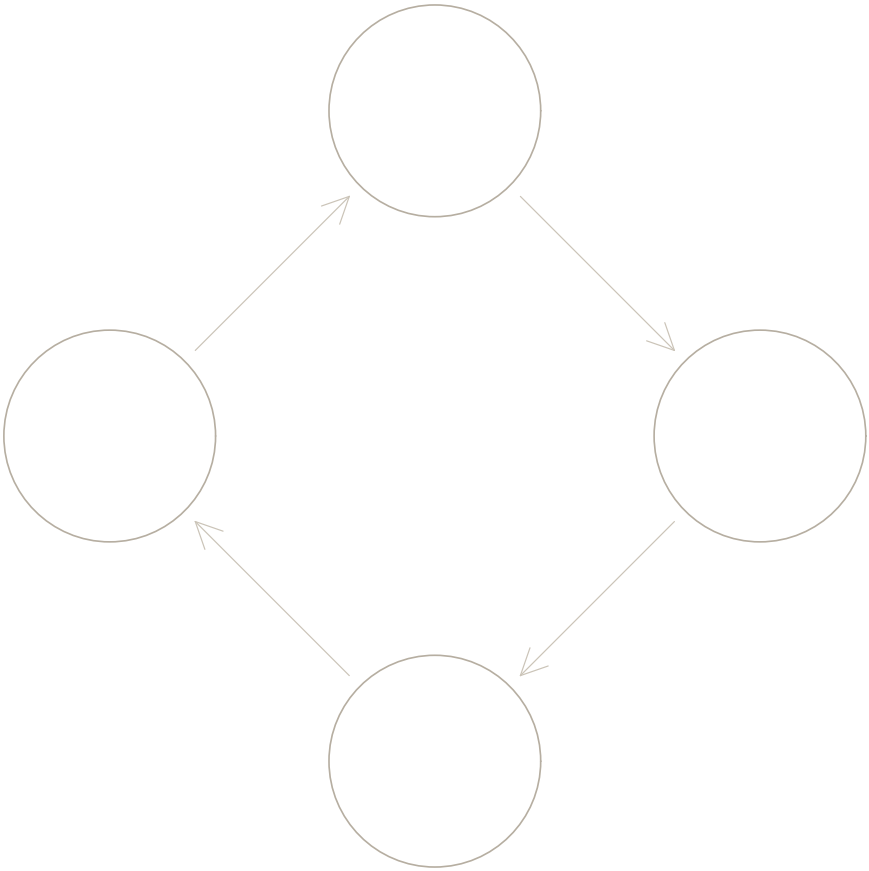Four-stage cycle